

Math Sec 1

1. Let the mean and variance of 7 observations 2, 4, 10, x , 12, 14, y , $x > y$, be 8 and 16 respectively. Two numbers are chosen from $\{1, 2, 3, x-4, y, 5\}$ one after another without replacement, then the probability, that the smaller number among the two chosen numbers is less than 4, is :
 (A) $\frac{2}{5}$ (B) $\frac{1}{3}$
 (C) $\frac{4}{5}$ (D) $\frac{3}{5}$
2. Let (α, β, γ) be the coordinates of the foot of the perpendicular drawn from the point $(5, 4, 2)$ on the line $\vec{r} = (-\hat{i} + 3\hat{j} + \hat{k}) + \lambda(2\hat{i} + 3\hat{j} - \hat{k})$. Then the length of the projection of the vector $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ on the vector $6\hat{i} + 2\hat{j} + 3\hat{k}$ is:
 (A) 3 (B) 4
 (C) $\frac{18}{7}$ (D) $\frac{15}{7}$
3. Let $\vec{c}$ and $\vec{d}$ be vectors such that $|\vec{c} + \vec{d}| = \sqrt{29}$ and $\vec{c} \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = (2\hat{i} + 3\hat{j} + 4\hat{k}) \times \vec{d}$. If $\lambda_1, \lambda_2 (\lambda_1 > \lambda_2)$ are the possible values of $(\vec{c} + \vec{d}) \cdot (-7\hat{i} + 2\hat{j} + 3\hat{k})$, then the equation $K^2x^2 + (K^2 - 5K + \lambda_1)xy + (3K + \frac{\lambda_2}{2})y^2 - 8x + 12y + \lambda_2 = 0$ represents a circle, for K equal to :
 (A) -1 (B) 1 (C) 4 (D) 2
4. The area of the region, inside the ellipse $x^2 + 4y^2 = 4$ and outside the region bounded by the curves $y = |x| - 1$ and $y = 1 - |x|$, is :
 (A) $2(\pi - 1)$ (B) $2\pi - \frac{1}{2}$
 (C) $3(\pi - 1)$ (D) $2\pi - 1$
5. If the coefficient of x in the expansion of $(ax^2 + bx + c)(1 - 2x)^{26}$ is -56 and the coefficients of x^2 and x^3 are both zero, then $a + b + c$ is equal to:
 (A) 1483 (B) 1300
 (C) 1500 (D) 1403
6. The number of strictly increasing functions f from the set $\{1, 2, 3, 4, 5, 6\}$ to the set $\{1, 2, 3, \dots, 9\}$ such that $f(i) \neq i$ for $1 \leq i \leq 6$, is equal to:
 (A) 21 (B) 27 (C) 28 (D) 22
7. Let O be the vertex of the parabola $x^2 = 4y$ and Q be any point on it. Let the locus of the point P, which divides the line segment OQ internally in the ratio 2 : 3 be the conic C. Then the equation of the chord of C, which is bisected at the point $(1, 2)$, is :
 (A) $4x - 5y + 6 = 0$ (B) $x - 2y + 3 = 0$
 (C) $5x - 4y + 3 = 0$ (D) $5x - y - 3 = 0$
8. Let the foci of a hyperbola coincide with the foci of the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$. If the eccentricity of the hyperbola is 5, then the length of its latus rectum is:
 (A) $24\sqrt{5}$ (B) 16
 (C) $\frac{96}{\sqrt{5}}$ (D) 12
9. Let PQ and MN be two straight lines touching the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ at the points A and B respectively. Let O be the centre of the circle and $\angle AOB = \pi/3$. Then the locus of the point of intersection of the lines PQ and MN is :
 (A) $x^2 + y^2 - 18x - 12y - 25 = 0$
 (B) $3(x^2 + y^2) - 18x - 12y + 25 = 0$
 (C) $3(x^2 + y^2) - 12x - 18y - 25 = 0$
 (D) $x^2 + y^2 - 12x - 18y - 25 = 0$
10. Let $f : \mathbf{R} \rightarrow (0, \infty)$ be a twice differentiable function such that $f(3) = 18$, $f'(3) = 0$ and $f''(3) = 4$. Then evaluate:

$$\lim_{x \rightarrow 1} (\log_e (\frac{f(2+x)}{f(3)})^{\frac{18}{(x-1)^2}})$$
 (A) 2 (B) 9 (C) 18 (D) 1
11. The value of $\operatorname{cosec} 10^\circ - \sqrt{3} \sec 10^\circ$ is equal to :
 (A) 6 (B) 8 (C) 4 (D) 2
12. The sum of all the roots of the equation $(x-1)^2 - 5|x-1| + 6 = 0$, is :
 (A) 5 (B) 4 (C) 1 (D) 3
13. The value of $\int_{-\pi/6}^{\pi/6} (\frac{\pi + 4x^{11}}{1 - \sin(|x| + \pi/6)}) dx$ is equal to :
 (A) 8π (B) 2π (C) 6π (D) 4π
14. Let $a_1, a_2, a_3, \dots$ be a G.P. of increasing positive terms such that $a_2 \cdot a_3 \cdot a_4 = 64$ and $a_1 + a_3 + a_5 = \frac{813}{7}$. Then $a_3 + a_5 + a_7$ is equal to:
 (A) 3244 (B) 3256 (C) 3248 (D) 3252
15. Let $y = y(x)$ be the solution curve of the differential equation $(1 + x^2) dy + (y - \tan^{-1} x) dx = 0$, with initial condition $y(0) = 1$. Then find the value of $y(1)$.
 (A) $\frac{2}{e^{\pi/4}} + \frac{\pi}{4} - 1$
 (B) $\frac{2}{e^{\pi/4}} - \frac{\pi}{4} - 1$
 (C) $\frac{4}{e^{\pi/4}} - \frac{\pi}{2} - 1$
 (D) $\frac{4}{e^{\pi/4}} + \frac{\pi}{2} - 1$

16. If the domain of the function $f(x) = \cos^{-1}\left(\frac{2x-5}{11-3x}\right) + \sin^{-1}(2x^2 - 3x + 1)$ is the interval $[\alpha, \beta]$, then $\alpha + 2\beta$ is equal to:
 (A) 1 (B) 2 (C) 5 (D) 3
17. Let a point A lie between the parallel lines L_1 and L_2 such that its distances from L_1 and L_2 are 6 and 3 units, respectively. Then the area (in sq. units) of the equilateral triangle ABC , where the points B and C lie on the lines L_1 and L_2 , respectively, is:
 (A) $12\sqrt{2}$ (B) $21\sqrt{3}$
 (C) 27 (D) $15\sqrt{6}$
18. If $x^2 + x + 1 = 0$, then the value of $(x + \frac{1}{x})^4 + (x^2 + \frac{1}{x^2})^4 + (x^3 + \frac{1}{x^3})^4 + \dots + (x^{25} + \frac{1}{x^{25}})^4$ is :
 (A) 145 (B) 175
 (C) 128 (D) 162
19. The number of relations, defined on the set $\{a, b, c, d\}$, which are both reflexive and symmetric, is equal to:
 (A) 16 (B) 256 (C) 64 (D) 1024
20. Let $\vec{a} = -\hat{i} + 2\hat{j} + 2\hat{k}$, $\vec{b} = 8\hat{i} + 7\hat{j} - 3\hat{k}$ and $\vec{c}$ be a vector such that $\vec{a} \times \vec{c} = \vec{b}$. If $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 4$, then $|\vec{d} + \vec{c}|^2$ is equal to :
 (A) 33 (B) 30 (C) 35 (D) 27

Math Sec 2

21. Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be a twice differentiable function such that the quadratic equation $f(x)m^2 - 2f'(x)m + f''(x) = 0$ in m , has two equal roots for every $x \in \mathbf{R}$. If $f(0) = 1$, $f'(0) = 2$, and (α, β) is the largest interval in which the function $f(\log_e x - x)$ is increasing, then $\alpha + \beta$ is equal to
22. Let $a_1 = 1$ and for $n \geq 1$, $a_{n+1} = \frac{1}{2}a_n + \frac{n^2 - 2n - 1}{n^2(n+1)^2}$. Then $|\sum_{n=1}^{\infty} (a_n - \frac{2}{n^2})|$ is equal to ____.
23. For some $\alpha, \beta \in \mathbf{R}$, let $A = \begin{bmatrix} \alpha & 2 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & \beta \end{bmatrix}$ be such that $A^2 - 4A + 2I = B^2 - 3B + I = O$. Then $(\det(\text{adj}(A^3 - B^3)))^2$ is equal to ____.
24. $6 \int_0^{\pi} |(\sin 3x + \sin 2x + \sin x)| dx$ is equal to ____.
25. Let $S = \{(m, n) : m, n \in \{1, 2, 3, \dots, 50\}\}$. If the number of elements (m, n) in S such that $6^m + 9^n$ is a multiple of 5 is p and the number of elements (m, n) in S such that $m + n$ is a square of a prime number is q , then $p + q$ is equal to ____.

Phy Sec 1

26. Initially a satellite of 100 kg is in a circular orbit of radius $1.5R_E$. This satellite can be moved to a circular orbit of radius $3R_E$ by supplying $\alpha \times 10^6$ J of energy. The value of α is _____. (Take Radius of Earth $R_E = 6 \times 10^6$ m and $g = 10$ m/s²)
 (A) 1000 (B) 150 (C) 500 (D) 100
27. If an alpha particle with energy 7.7 MeV is bombarded on a thin gold foil, the closest distance from nucleus it can reach is ____ m. (Atomic number of gold = 79 and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ in SI units)
 (A) 2.95×10^{-14} (B) 2.95×10^{-16}
 (C) 3.85×10^{-14} (D) 3.85×10^{-16}
28. The electric field in a plane electromagnetic wave is given by :
 $E_y = 69 \sin [0.6 \times 10^3 x - 1.8 \times 10^{11} t]$ V/m.
 The expression for magnetic field associated with this electromagnetic wave is ____ T.
 (A) $B_z = 2.3 \times 10^{-7} \sin [0.6 \times 10^3 x + 1.8 \times 10^{11} t]$
 (B) $B_z = 2.3 \times 10^{-7} \sin [0.6 \times 10^3 x - 1.8 \times 10^{11} t]$
 (C) $B_y = 69 \sin [0.6 \times 10^3 x + 1.8 \times 10^{11} t]$
 (D) $B_y = 2.3 \times 10^{-7} \sin [0.6 \times 10^3 x - 1.8 \times 10^{11} t]$
29. Consider a modified Bernoulli equation.
 $(P + \frac{A}{Bt^2}) + \rho g(h + Bt) + \frac{1}{2}\rho V^2 = \text{constant}$
 If t has the dimension of time then the dimensions of A and B are _____, _____ respectively.
 (A) $[ML^0 T^{-1}]$ and $[M^0 LT]$
 (B) $[ML^0 T^{-2}]$ and $[M^0 LT^{-1}]$
 (C) $[ML^0 T^{-1}]$ and $[M^0 LT^{-1}]$
 (D) $[ML^0 T^{-2}]$ and $[M^0 LT^{-2}]$
30. A 4 kg mass moves under the influence of a force $\vec{F} = (4t^3\hat{i} - 3t\hat{j})$ N where t is the time in second.
 If mass starts from origin at $t = 0$, the velocity and position after $t = 2$ s will be :
 (A) $\vec{v} = 4\hat{i} - \frac{3}{2}\hat{j}$ $\vec{r} = \frac{8}{5}\hat{i} - \hat{j}$
 (B) $\vec{v} = 4\hat{i} - \frac{3}{2}\hat{j}$ $\vec{r} = \frac{6}{5}\hat{i} - \hat{j}$
 (C) $\vec{v} = 4\hat{i} + \frac{5}{2}\hat{j}$ $\vec{r} = \frac{8}{5}\hat{i} + 2\hat{j}$